Network Threat Analysis & Detection Report

1. Beaconing Behaviours (C2-like Pattern)

→ One of the most notable findings was the presence of regular outbound connections to the same external IP address at consistent intervals (~14–20 seconds).

→ This pattern closely resembles **beaconing behavior**, where a system periodically communicates with a remote server. The low variance in timing suggests automated communication rather than user-driven activity.

→ Conclusion :-

While no actual malware was used, this behavior is consistent with techniques used by command-and-control (C2) frameworks such as Cobalt Strike, where infected hosts regularly check in with a server.

2. DNS + TLS Communication pattern

→ I observed a recurring sequence where:

DNS queries were made to resolve domains
Followed by outbound TLS (HTTPS) connections

→ This “resolve → connect” pattern is commonly used in staged communication models, especially by loaders and initial access malware.

→ Conclusion :-

This behavior is **similar to communication patterns seen in malware like Emotet**, where DNS is used to locate infrastructure before establishing encrypted connections.

3. Mixed HTTP and DNS Activity

→ Some HTTP traffic was present
→ However, it lacked detailed metadata such as user-agent strings
→ DNS activity was also observed alongside it

→ The absence of typical HTTP characteristics suggests that the traffic may not be generated by a standard browser, but rather by automated processes.

→ Conclusion :-

This pattern resembles **flexible communication techniques used by malware such as QakBot**, which often combines multiple protocols (HTTP + DNS) for resilience.

4. DNS Activity Analysis

→ DNS queries included:

- `beam.scs.splunk.com`
- Reverse DNS lookups (PTR records)

→ There were:

- No unusually long domains
- No encoded or high-entropy subdomains

No excessive number of unique queries

→ Typical DNS tunneling involves long, random-looking domains and a high number of unique queries. These indicators were not present.

→ Conclusion :-

The DNS traffic appears **benign and does not indicate tunneling or data exfiltration**.

→ Correlation Across Tools

- **Suricata** detected protocol-level activity (DNS, TLS, ICMP)
- **Zeek** provided detailed behavioral logs (connections, DNS, TLS)
- **Splunk** was used to search, correlate, and visualize events

By combining these tools, I was able to:

- Identify patterns (e.g., beaconing)
- Validate observations across multiple data sources

Perform basic threat hunting using real logs

→ Key Findings

- Identified **beacon-like communication patterns** with regular intervals
- Observed **DNS followed by TLS communication**, indicating staged behavior
- Detected **encrypted outbound traffic with limited visibility**
- Confirmed **no DNS tunneling or data exfiltration patterns**
- Successfully correlated Suricata alerts with Zeek logs using Splunk

→ Final Conclusion

This lab demonstrates how even simulated traffic can exhibit patterns similar to real-world attack techniques.

Rather than identifying specific malware, the focus was on **detecting behaviors** such as beaconing, staged communication, and protocol misuse. These behavious are commonly associated with command-and-control activity in real environments.

Overall, this project helped develop practical skills in:

- Network monitoring
- Traffic analysis
- Threat detection

```
analyst@SOC:/var/log/suricata$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for analyst:
04/26/2026-18:25:05.332325  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:40270 -> 34.107.243.93:443
04/26/2026-18:25:05.332531  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 34.107.243.93:443 -> 10.0.2.15:40270
04/26/2026-18:25:05.372217  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 34.107.243.93:443 -> 10.0.2.15:40270
04/26/2026-18:25:05.413426  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:40270 -> 34.107.243.93:443
04/26/2026-18:25:06.332606  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:48422 -> 151.101.193.91:443
04/26/2026-18:25:06.332777  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.193.91:443 -> 10.0.2.15:48422
04/26/2026-18:25:06.355086  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 151.101.193.91:443 -> 10.0.2.15:48422
04/26/2026-18:25:06.355111  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:48422 -> 151.101.193.91:443
04/26/2026-18:25:06.969274  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:34420
04/26/2026-18:25:07.928723  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:35274 -> 34.107.243.93:443
04/26/2026-18:25:13.977304  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {UDP} 10.0.2.15:56340 -> 8.8.8.8:53
04/26/2026-18:25:13.977462  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {UDP} 10.0.2.15:54767 -> 8.8.8.8:53
04/26/2026-18:25:14.022551  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {UDP} 8.8.8.8:53 -> 10.0.2.15:56340
04/26/2026-18:25:14.020594  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {UDP} 8.8.8.8:53 -> 10.0.2.15:54767
04/26/2026-18:25:14.021856  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 8.8.8.8:53 -> 10.0.2.15:36184
04/26/2026-18:25:14.063999  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 8.8.8.8:53 -> 10.0.2.15:36184
04/26/2026-18:25:14.064042  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:36184 -> 8.8.8.8:53
04/26/2026-18:25:14.071142  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.119002  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.119031  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.119206  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.119211  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.119405  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.119411  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.119622  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.119628  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.120209  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.120337  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.166369  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.167276  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.170309  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.214711  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.216803  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:14.217263  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.266923  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.90:443 -> 10.0.2.15:55548
04/26/2026-18:25:14.266993  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:55548 -> 54.192.142.90:443
04/26/2026-18:25:16.061030  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 8.8.8.8:53 -> 10.0.2.15:36184
04/26/2026-18:25:16.061090  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:36184 -> 8.8.8.8:53
04/26/2026-18:25:16.061203  [**] [1:1000003:1] DNS QUERY DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 8.8.8.8:53 -> 10.0.2.15:36184
04/26/2026-18:25:17.948160  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 54.192.142.116:443 -> 10.0.2.15:50780
04/26/2026-18:25:26.428454  [**] [1:1000004:1] TLS CONNECTION DETECTED [**] [Classification: (null)] [Priority: 3] {TCP} 10.0.2.15:35664 -> 151.101.65.91:443
```

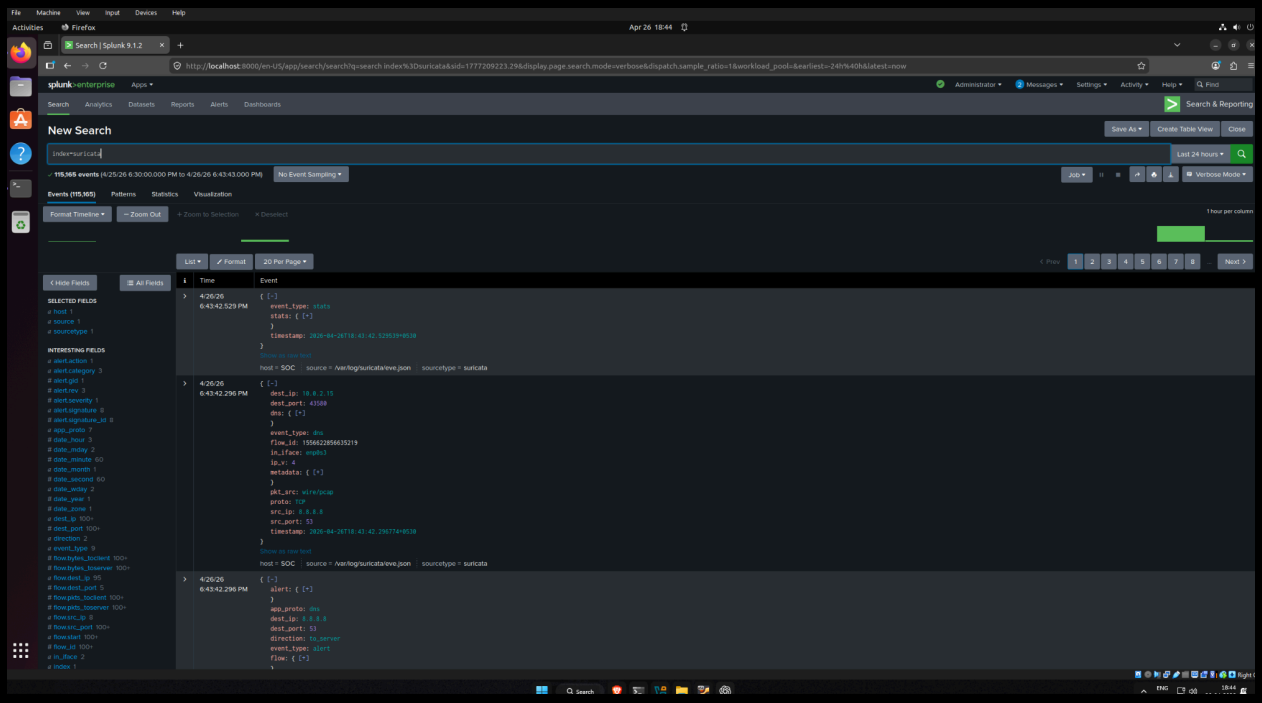
→ Suricata alerts showing detected DNS and TLS activity from analyzed traffic.

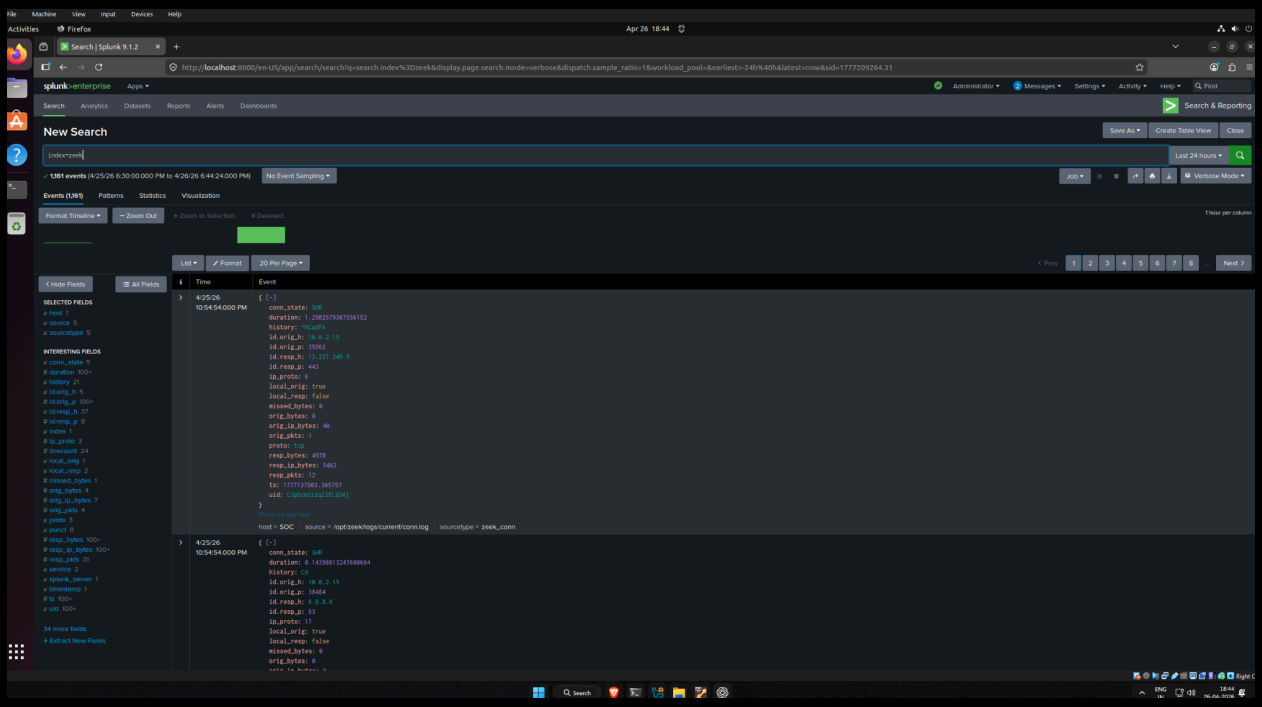
```
root@SOC:/opt/zeek/logs/current# cat conn.log | python3 -c "
import sys,json
from collections import defaultdict

conns=defaultdict(list)

for line in sys.stdin:
    try:
        j=json.loads(line)
        conns[j['id.resp_h']].append(float(j['ts']))
    except:
        pass

for ip,times in conns.items():
    if len(times)>5:
        times=sorted(times)
        intervals=[times[i+1]-times[i] for i in range(len(times)-1)]
        avg=sum(intervals)/len(intervals)
        print(f'{ip} -> connections:{len(times)} avg_interval:{avg:.2f}')
"
8.8.8.8 -> connections:509 avg_interval:2.22
13.227.249.46 -> connections:44 avg_interval:25.48
13.227.249.117 -> connections:34 avg_interval:30.96
13.227.249.54 -> connections:44 avg_interval:25.14
13.227.249.9 -> connections:38 avg_interval:28.40
34.160.144.191 -> connections:10 avg_interval:0.09
34.111.211.209 -> connections:12 avg_interval:0.01
151.101.1.91 -> connections:7 avg_interval:50.01
34.120.237.76 -> connections:12 avg_interval:0.01
34.223.169.39 -> connections:10 avg_interval:12.27
107.21.114.30 -> connections:9 avg_interval:45.10
root@SOC:/opt/zeek/logs/current#
```

→ Zeek conn.log output demonstrating repeated connections to external IP addresses with semi-regular intervals, indicating potential beaconing activity.

```
root@SOC:/opt/zeek/logs/current# cat dns.log | python3 -c "
import sys,json
from collections import Counter

domains=Counter()

for line in sys.stdin:
    try:
        j=json.loads(line)
        domains[j.get('query','unknown')] += 1
    except:
        pass

for d,c in domains.most_common(20):
    print(c,d)
"
264 unknown
97 beam.scs.splunk.com
16 connectivity-check.ubuntu.com
12 quickdraw.splunk.com
9 www.bbc.com
9 docs.splunk.com
8 telemetry-splkmobile.dataeng.splunk.com
8 ads.mozilla.org
8 mozilla-ohttp.fastly-edge.com
8 www.deccanherald.com
7 www.theverge.com
7 m.economictimes.com
6 support.mozilla.org
6 www.youtube.com
6 www.thecut.com
6 www.mozilla.org
6 img-getpocket.cdn.mozilla.net
6 detectportal.firefox.com
6 www.wikipedia.org
6 idp.login.splunk.com
root@SOC:/opt/zeek/logs/current#
```

→ DNS analysis revealed standard domain resolution activity including queries to public services and reverse lookups. No high-entropy domains or excessive unique queries were observed, indicating no evidence of DNS tunneling.





→ Correlation Across Tools